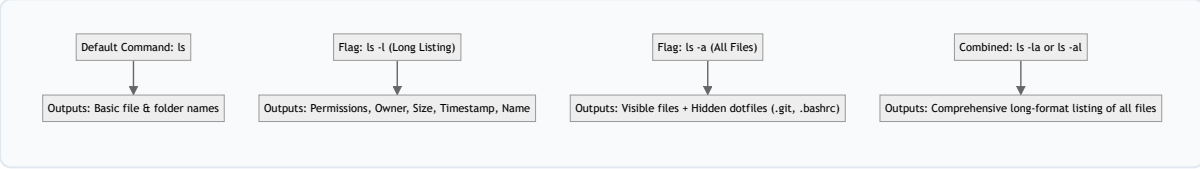
Submodule 34.09: Mastering Command Flags: Manual Pages (`man`), `ls -l`, and `ls -a`

Topic Overview & The Concept of Flags

In Unix-like shells, **Flags** (also termed **options** or **switches**) are special tokens passed after a command name to alter its default behavior or request additional metadata. Flags typically begin with a single hyphen (`-`) for single-character options or double hyphens (`--`) for verbose word options.



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1. System Documentation via `man` (Manual Pages)

Every standard Unix command possesses a built-in instruction manual embedded in the operating system:

```
# Read the official manual page for ls
man ls
```

Navigating the `man` Viewer:

- Use the **Up/Down Arrow keys** or **Spacebar** to scroll through documentation.
- **Search**: Type `/keyword` and press Enter.
- **Quit / Exit**: Press the `q` key on your keyboard to instantly return to the command prompt.

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2. Essential Flags for the `ls` Command

A. Long Listing Format (`ls -l`)

By default, `ls` outputs only names. Adding `-l` prints detailed metadata in tabular format:

```
ls -l
# Output example:
# drwxr-xr-x  4 Shraddha  staff   128 Aug 31 14:20 Delta
# -rw-r--r--  1 Shraddha  staff  2048 Aug 31 15:10 README.md
```

Anatomy of a Long Listing Entry:

File Permissions	Links	Owner	Group	Size (Bytes)	Date / T
-rw-r–r–	1	Shraddha	staff	2048	Aug 31 15

- First character: `d` indicates a directory; `-` indicates a regular file.
- `rwxr-xr-x` : Read (`r`), write (`w`), execute (`x`) permissions for user, group, and others.
- File size is displayed in raw bytes.

B. All Files Including Hidden Dotfiles (`ls -a`)

In Unix systems, any file whose name begins with a period (`.`) is considered a **hidden file** (e.g., `.bashrc` , `.zshrc` , `.git` , `.gitignore` , `.env`). These are excluded from default `ls` output.

```
ls -a
# Output:
# .  ..  .bash_profile  .git  .gitignore  Desktop  Documents
```

- `.` : Represents the current directory itself.
- `..` : Represents the parent directory.

C. Combining Flags (`ls -la` or `ls -al`)

Flags can be chained together in any order under a single hyphen:

```
ls -la
# Displays full metadata for all files, including hidden dotfiles
```